

Student Performance, Mentorship & Employability Platform Requirements Document

This document captures all requirements provided by the stakeholder without omitting any requested functionality. It is intended as a feature requirements document and foundation for a full PRD.

1. Administrator Module

- Upload students using Excel containing Name, PRN, Official Email, Phone, Batch, Division and Current Semester.
- Automatically create user accounts after import.
- Automatically generate temporary passwords using configurable patterns (e.g. firstname + birth year).
- Manage students individually after import.
- Manage list of teachers and map them to departments such as CSE, ECE, etc.
- Manage departments, programs, batches, semesters and divisions.
- Manage subjects for every batch/year. Support Excel upload.
- Subject fields: Semester, Course Code, Course Name, Marking Scheme.
- Group subjects into competency domains for analytics. Example: DSA (Semester 3) + Competitive Programming (Semester 4) belong to DSA domain; Java belongs to Backend/OOP; Cloud Computing belongs to Cloud/Web Infrastructure; BML and Deep Learning belong to AI/ML.
- Allow one subject to belong to multiple domains with configurable weights.
- Manage university-wide marking schemes such as TL3.
- Example TL3: CCA1 (10), Mid Term (25), CCA2 (10), LCA1 (5), LCA2 (5), LCA3 (5).
- Support actual marks and reduced marks separately because examinations may be conducted for different marks than finally considered.
- Upload student marks through Excel: PRN, Course Code, Exam Name, Marks Obtained (Actual), Actual Maximum Marks.
- Future: automatically normalize marks and calculate GPA/CGPA.
- For now, allow administrators to upload semester-wise GPA and CGPA manually.
- Bulk promote students to the next semester.
- Allow excluding selected students during promotion because of backlog or year down.
- Maintain audit history of all administrative actions.

2. Student Module

- First login requires password change and account activation.

- Student completes profile after first login.
- Profile fields: About Me, Professional Photograph, Best Five Subjects of Class X.
- Students can add skills grouped by domains such as AI/ML, Cybersecurity, DSA/Backend, Planning and Other.
- Select existing skills or create custom skills.
- Administrators manage master list of domains and skills.
- Skill proficiency uses Likert scale (Novice, Beginner, Intermediate, Advanced, Expert).
- For every skill, attach related certificate directly from the same screen (creates linked certificate automatically).
- For every skill, link related project without recreating the project.
- Add communication languages and soft skills.
- Students specify desired skills they want to learn to help mentors recommend learning paths.

3. Certificate Module

- Certificate Name
- Description
- File Upload
- Certified By
- Related Skill
- Uploading a certificate automatically creates a dashboard card/placeholder.
- Future visibility controls for recruiters and mentors.

4. Project Module

- Project Name
- Description
- Project Image
- Project Link
- GitHub Link
- Select Domains
- Select Related Skills
- Choose College Mentors (Optional).
- Visibility option: Show on Recruiter Profile or keep Mentor Only.

5. Mentor Module

- Automatically display all current semester subject teachers.
- All subject teachers can monitor the student's academic progress.
- Student can additionally choose preferred mentors.
- Mentors should view academics, skills, projects, certificates, internships and experiences.
- Future: mentor notes, action items and meeting history.

6. Experience Module

- Internship: Paid/Unpaid, Company Name, Company Mentor (optional), College Mentor, Platform (Online/Offline/Hybrid), Feedback, Start Date, End Date, Visibility.
- Jobs: Complete employment details with visibility option.
- Co-curricular activities: Mandatory recording of volunteering, clubs, events, organizing committees and responsibilities so the institution maintains a record.
- Recommended additions: Hackathons, Research Papers, Workshops, Competitions, Leadership Positions, Awards and Open Source Contributions.

7. Teacher Dashboard

- Navigation hierarchy: Institution → Polytechnic & Skill Development → Department → Year → Division → Student.
- Teachers can view analytics for entire department or only students who selected them as mentors.
- Subject-wise analysis.
- Exam-wise analysis.
- Show students appeared, passed, failed, highest, lowest, average and students scoring above 60%.
- CCA1 → Mid Term → CCA2 progression charts.
- Overall subject comparison.
- Student performance across semesters to identify improvement or decline.
- Performance across competency domains (AI, Web, DSA, Cloud, etc.).
- Individual student dashboard and batch/division dashboard.

8. Recommended Analytics & Graphs

- Subject-wise average marks.
- Pass/Fail distribution.
- Highest/Lowest marks.
- Marks distribution histogram.
- CCA1 vs Mid Term vs CCA2 trend.
- Semester GPA trend.
- CGPA progression.
- Domain-wise radar chart.
- Student percentile ranking.
- Heatmap of students vs subjects.
- Top performers.
- At-risk students.
- Improvement index.
- Consistency index.
- Class average vs student comparison.
- Domain contribution chart.
- Certification growth.

- Project count by domain.
- Internship statistics by company/domain.
- Placement readiness score.
- Skill distribution by department.
- Students without internships.
- Students without projects.
- Students without certifications.
- Mentor workload.
- Department comparison.
- Batch comparison.
- Division comparison.

9. Skill Analytics & Data-driven Mentorship

- View skills student-wise or skill-wise.
- Search students by a specific skill (e.g., Mobile App Development).
- View all certifications linked to each skill.
- View projects linked to each skill.
- Identify students lacking skills in particular domains.
- Identify best students per domain.
- Recommend certifications, hackathons, projects and learning resources.
- Track internship participation by company and domain.
- Identify placement-ready students.
- Provide mentors with data-driven recommendations for each student.

10. Employer Portal

- Generate secure share links.
- Share individual students.
- Share selected custom student groups.
- Share entire divisions, batches, years or departments.
- Display resume, skills, projects, certifications, internships, achievements, GitHub and portfolio.
- Display overall academic performance summary without exposing detailed subject marks unless permitted.
- Provide recruiter-friendly profile pages.